

Incident Response Report – Task 2

(SIEM Analysis)

Introduction

This report documents the analysis of simulated security events using a SIEM tool as part of Task 2 in the Future Interns Cybersecurity Internship. The objective was to monitor logs, identify suspicious activities, classify alerts, and recommend response actions.

SIEM Tool Used

Tool: Elastic Stack (ELK)

Module Used: Kibana – Discover

Data Source: Simulated security logs

Detected Security Events

1. Event Type : Malware Detected

Description : Multiple malware alerts detected across endpoints

The screenshot shows the Kibana Discover interface with the following details:

- Search Bar:** Contains the query `attachment.content : "malware detected"`.
- Left Sidebar:**
 - Selected fields:** 1 field selected: `attachment.content`.
 - Popular fields:** 9 fields listed: `attachment.content`, `_id`, `_ignored`, `_index`, `_score`, `attachment.content_length`, `attachment.content_type`, `attachment.format`, `attachment.language`.
 - Available fields:** 4 fields listed: `attachment.content`, `attachment.content_length`, `attachment.content_type`, and an `Add a field` button.
- Main Panel:** Displays a list of events. The first event is highlighted with a yellow box and contains the following fields:
 - `ip=10.0.0.5`
 - `action=malware_detected`
 - `threat=Trojan Detected`
 - `timestamp=2025-07-03 08:30:14`
 - `user=eve`
 - `ip=172.16.0.3`
 - `action=login success`
 - `timestamp=2025-07-03 08:21:14`
 - `user=david`
 - `ip=172.16.0.3`
 - `action=connection attempt`
 - `timestamp=2025-07-03 05:45:14`
 - `user=david`
 - `ip=172.16.0.3`
 - `action=malware_detected`
 - `threat=Trojan Detected`
 - `timestamp=2025-07-03 08:00:14`
 - `user=alice`
 - `ip=198.51.100.42`
 - `action=login success`
 - `timestamp=2025-07-03 04:19:14`
 - `user=alice`
 - `ip=198.51.100.42`
 - `action=malware_detected`
 - `threat=Rootkit Signature`
 - `timestamp=2025-07-03 05:30:14`
 - `user=eve`
 - `ip=192.168.1.101`
 - `action=malware_detected`
 - `threat=Trojan Detected`
 - `timestamp=2025-07-03 06:10:14`
 - `user=david`
 - `ip=203.0.113.77`
 - `action=file accessed`
 - `timestamp=2025-07-03 05:42:14`
 - `user=eve`
 - `ip=203.0.113.77`
 - `action=malware_detected`
 - `threat=Trojan Detected`
 - `timestamp=2025-07-03 07:02:14`
 - `user=alice`
 - `ip=203.0.113.77`
 - `action=login failed`
 - `timestamp=2025-07-03 04:18:14`
 - `user=bob`
 - `ip=198.51.100.42`
 - `action=login success`
 - `timestamp=2025-07-03 09:02:14`
 - `user=david`
 - `ip=203.0.113.77`
 - `action=login failed`
 - `timestamp=2025-07-03 09:07:14`
 - `user=eve`
 - `ip=203.0.113.77`
 - `action=login success`
 - `timestamp=2025-07-03 04:47:14`
 - `user=bob`
 - `ip=10.0.0.5`
 - `action=login failed`
 - `timestamp=2025-07-03 07:38:14`
 - `user=charlie`
 - `ip=172.16.0.3`
 - `action=connection attempt`
 - `timestamp=2025-07-03 07:57:14`
 - `user=david`
 - `ip=10.0.0.5`
 - `action=file accessed`
 - `timestamp=2025-07-03 07:44:14`
 - `user=bob`
 - `ip=203.0.113.77`
 - `action=connection attempt`
 - `timestamp=2025-07-03 05:33:14`
 - `user=david`
 - `ip=198.51.100.42`
 - `action=file accessed`
 - `timestamp=2025-07-03 04:19:14`
 - `user=david`
 - `ip=10.0.0.5`
 - `action=connection attempt`
 - `timestamp=2025-07-03 04:29:14`
 - `user=alice`
 - `ip=192.168.1.101`
 - `action=malware_detected`
 - `threat=Trojan Detected`
 - `timestamp=2025-07-03 07:51:14`
 - `user=eve`
 - `ip=10.0.0.5`
 - `action=malware_detected`
 - `threat=Rootkit Signature`
 - `timestamp=2025-07-03 04:53:14`
 - `user=david`
 - `ip=203.0.113.77`
 - `action=login success`
 - `timestamp=2025-07-03 04:23:14`
 - `user=charlie`
 - `ip=198.51.100.42`
 - `action=login failed`
 - `timestamp=2025-07-03 05:27:14`

2. Event Type : Login Failed

Description : Repeated failed login attempts indicating brute-force behavior

Deployments Discover

Try ES|QL Inspect Alerts + Save

Untitled +

Data view soc-tak2 attachment.content : "login failed" Refresh

Search field names 0

Documents (1) Field statistics

Columns 1 Sort fields

Selected fields 1

- attachment.content

Popular fields 9

- attachment.content
- _id
- _ignored
- _index
- _score
- attachment.content_length
- attachment.content_type
- attachment.format
- attachment.language

Available fields 4

- attachment.content
- attachment.content_length
- attachment.content_type

Add a field

attachment.content

```
threat=Trojan Detected 2025-07-03 07:02:14 | user=alice | ip=203.0.113.77 | action=login_failed 2025-07-03 04:18:14 | user=bob | ip=198.51.100.42 | action=login success 2025-07-03 09:02:14 | user=david | ip=203.0.113.77 | action=login_failed 2025-07-03 09:07:14 | user=eve | ip=203.0.113.77 | action=login success 2025-07-03 04:47:14 | user=bob | ip=10.0.0.5 | action=login_failed 2025-07-03 07:38:14 | user=charlie | ip=172.16.0.3 | action=connection attempt 2025-07-03 07:57:14 | user=david | ip=10.0.0.5 | action=file accessed 2025-07-03 07:44:14 | user=bob | ip=203.0.113.77 | action=connection attempt 2025-07-03 05:33:14 | user=david | ip=198.51.100.42 | action=file accessed 2025-07-03 04:19:14 | user=david | ip=10.0.0.5 | action=connection attempt 2025-07-03 04:29:14 | user=alice | ip=192.168.1.101 | action=malware detected | threat=Trojan Detected 2025-07-03 07:51:14 | user=eve | ip=10.0.0.5 | action=malware detected | threat=Rootkit Signature 2025-07-03 04:53:14 | user=david | ip=203.0.113.77 | action=login success 2025-07-03 04:23:14 | user=charlie | ip=198.51.100.42 | action=login_failed 2025-07-03 05:27:14 | user=david | ip=203.0.113.77 | action=connection attempt 2025-07-03 07:46:14 | user=bob | ip=10.0.0.5 | action=login success 2025-07-03 04:41:14 | user=alice | ip=172.16.0.3 | action=malware detected | threat=Spyware Alert 2025-07-03 09:10:14 | user=bob | ip=198.51.100.42 | action=file accessed 2025-07-03 07:36:14 | user=david | ip=10.0.0.5 | action=connection attempt 2025-07-03 08:31:14 | user=eve | ip=203.0.113.77 | action=file accessed 2025-07-03 05:49:14 | user=charlie | ip=192.168.1.101 | action=connection attempt 2025-07-03 06:21:14 | user=alice | ip=203.0.113.77 | action=login success 2025-07-03 07:44:14 | user=bob | ip=192.168.1.101 | action=connection attempt 2025-07-03 04:23:14 | user=bob | ip=172.16.0.3 | action=login_failed
```

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3. Event Type : Trojan Detected

Description : Trojan signatures found in system logs

Deployments Discover

Try ES|QL Inspect Alerts + Save

Untitled +

Data view soc-tak2 attachment.content : "trojan detected" Refresh

Search field names 0

Documents (1) Field statistics

Columns 1 Sort fields

Selected fields 1

- attachment.content

Popular fields 9

- attachment.content
- _id
- _ignored
- _index
- _score
- attachment.content_length
- attachment.content_type
- attachment.format
- attachment.language

Available fields 4

- attachment.content
- attachment.content_length
- attachment.content_type

Add a field

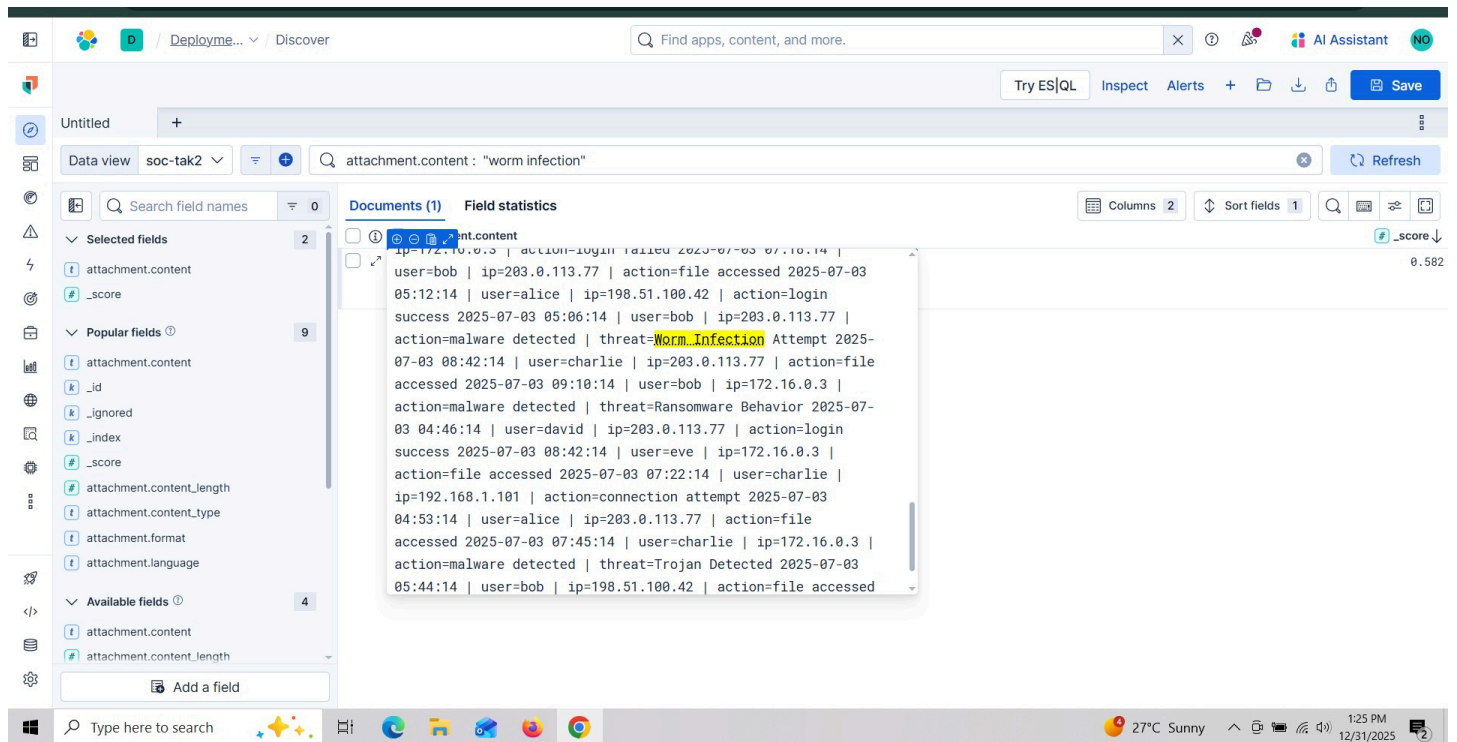
attachment.content

```
success 2025-07-03 04:27:14 | user=david | ip=172.16.0.3 | action=connection attempt 2025-07-03 05:40:14 | user=bob | ip=10.0.0.5 | action=malware detected | threat=Trojan Detected 2025-07-03 08:30:14 | user=eve | ip=172.16.0.3 | action=login success 2025-07-03 08:21:14 | user=david | ip=172.16.0.3 | action=connection attempt 2025-07-03 05:45:14 | user=david | ip=172.16.0.3 | action=malware detected | threat=Trojan Detected 2025-07-03 08:00:14 | user=alice | ip=198.51.100.42 | action=login success 2025-07-03 04:19:14 | user=alice | ip=198.51.100.42 | action=malware detected | threat=Rootkit Signature 2025-07-03 05:30:14 | user=eve | ip=192.168.1.101 | action=malware detected | threat=Trojan Detected 2025-07-03 06:10:14 | user=david | ip=203.0.113.77 | action=file accessed 2025-07-03 05:42:14 | user=eve | ip=203.0.113.77 | action=malware detected | threat=Trojan Detected 2025-07-03 07:02:14 | user=alice | ip=203.0.113.77 | action=login failed 2025-07-03 04:18:14 | user=bob | ip=198.51.100.42 | action=login success 2025-07-03 09:02:14 | user=david | ip=203.0.113.77 | action=login failed 2025-07-03 09:07:14 | user=eve | ip=203.0.113.77 | action=login success 2025-07-03 04:47:14 | user=bob | ip=10.0.0.5 | action=login failed 2025-07-03 07:38:14 | user=charlie | ip=172.16.0.3 | action=connection attempt 2025-07-03 07:57:14 | user=david | ip=10.0.0.5 | action=file accessed 2025-07-03 07:44:14 | user=bob | ip=203.0.113.77 | action=connection attempt 2025-07-03 05:33:14 | user=david | ip=198.51.100.42 | action=file accessed 2025-07-03 04:19:14 | user=david | ip=10.0.0.5 | action=connection attempt 2025-07-03 04:29:14 | user=alice | ip=192.168.1.101 | action=malware detected | threat=Trojan Detected 2025-07-03 07:51:14 | user=eve | ip=10.0.0.5 | action=malware detected | threat=Rootkit Signature 2025-07-03 04:53:14 | user=david | ip=203.0.113.77 | action=login success 2025-07-03 04:23:14 | user=charlie | ip=198.51.100.42 | action=login failed 2025-07-03 05:27:14 |
```

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4. Event Type : Worm Infection

Description : Worm infection attempts identified



Incident Analysis

- **Example:** Malware Detection
- **Action:** malware detected
- **Threat Type:** Trojan / Worm
- **Severity:** High
- **Affected Users:** alice, bob, eve
- **Source IPs:** Internal and external IPs observed

These events indicate possible endpoint compromise and unauthorized access attempts.

Alert Type	Severity	Reason
-----	-----	-----
Malware Detection	High	Active malicious activity
Trojan Detection	High	Known malware signature
Worm Infection	High	Self-propagating threat
Login Failed	Medium	Possible brute-force

Incident Response Actions

- Alerts were identified and reviewed using SIEM dashboard
- High-severity alerts were prioritised
- Incident categorized as Malware Infection
- Recommended escalation to Incident Response Team

Recommendations

- Isolate affected systems

- Run full antivirus scan
- Patch vulnerable systems
- Enforce strong authentication (MFA)
- Monitor login attempts continuously

Conclusion

This task provided hands-on experience in SOC operations, SIEM log analysis, alert classification, and incident response reporting.

